2 – August Focus Group – FG2

Interviewer

First of all, just to kind of get everyone introduced to each other, I'd just like if we kind of go around and kind of say what type one diabetes technology you're using at the moment in terms of the pump and CGM, what smartwatch you've got as part of the study and then a bit of type one diabetes technology that you have disliked and that could be in the past or current or yeah, any level there. So, I'll go off first. As I was kind of saying the beginning of this, I've recently moved on to the Tandem t:slim and the Dexcom G6 sensors. I'm having to self fund those at the moment just a bit of a pain, but I'm hoping in the next few months they're kind of sort themselves out and the NICE guidelines helps a little bit with that. Uhm, I actually have been using a Fitbit Sense for a few years now as part of the PhD. I kind of got one just to kind of get a feel for it and see what it could be used with. And then my bit of total diabetes technology, I'm not a fan of is. Well, I've got it here. So, this is the blood glucose metre from the spirit combo, which is kind of Bluetooth to the pump, the Accu-Chek one. And so it seemed really good when I was using it and for a long time. But there are times when you're kind of using it and every time you use it, it tries to connect to the pump, which takes about 15/20 seconds. Real pain, especially when you don't even need that, you just want to quickly check something so that kind of has grown into a pain, particularly now as I still use as my BM kit, don't use the pump and it's still trying to connect with a pump that you know doesn't really exist anymore. But yeah, so now I'll kind of go around. So, [P01], do you mind going next and just kind of going through that?

P01

Yeah, sure. Hi. So, I'm [P01]. I use the Medtronic pump and the FreeStyle Libre. The Fitbit that I've got is the Fitbit Luxe, I think. And in terms of diabetes technology that I don't like. Oh, I don't know, it's hard. They've all got their like frustrations, but I guess recently when the FreeStyle app decided to delete itself in the middle of the night, that was really annoying. So probably just when things like that happen.

Interviewer

Yeah, no, I heard a few people who went through that. It does sound, yeah, quite a scary moment you realised too. And I'm sure we'll pick it up at some point.

P01

Yeah, not good at 3:00 in the morning.

Interviewer

No, definitely. Uh yeah. Do you mind picking someone else to go next?

P01

Yeah, uh, [P05].

P05

Yeah. Hi. Yeah. I'm. I'm [P05]. So, I use the Medtronic 640 at pump at the minute, so I'm due at pump upgrade like in the next sort of six weeks to two months. So, we recently had like a show and tell day for all those. So obviously as [Interviewer] said as well like I'm hoping now with the changing guidelines with the CGM stuff that it's going to be a case of getting onto a closed-loop system hopefully really soon. My Fitbit is the Versa 4. Um and obviously I've had that since being involved with the study with [Interviewer]. Diabetes technology that I don't really like. There's a lot of bugs with the FreeStyle Libre app. Do you remember when it went through that phase of like constantly losing connection and it was like, “sensor error. Scan your sensor to reconnect. Scan your sensor to reconnect” and like it was happening, like maybe three or four times an hour when I was at work and I was like, “I can't”. Like I'm a teacher, I was like, “I can't have my phone constantly, like this should just be working”. I guess that's more of like an app problem than an actual diabetes problem, but obviously that facilitates us being diabetic. Um. So, I find that really annoying, but it has been better since and I like the fact that it's changed recently, so you don't actually have to scan, it just gives you automatic blood sugar updates. Which is good because you get like the notification bar and it's like blood sugar going high or low or whatever and it gives you your reading as well which is good.

Interviewer

Yeah, definitely. I think it's really cool they've managed to do that and without everybody having to pick up new technology, there's just kind of an upgrade to the app you download.

P05

Yeah, exactly. [P02], do you want to go next?

P02

Hi. Um. So, I'm [P02]. I'm currently using the Tandem t:slim with the Dexcom G6. And by the sound of things I'm super lucky because I have my own prescription, but my team and are really good and I think I got in. They move me onto the pump. Right when the G6 started to be available on prescription. So, I kind of scooted in on the end of that. So that's how I got mine. Diabetes technology I dislike. Kind of with you guys on the app thing. I I think before I was on before I was on the Dexcom, I was on the Libre FreeStyle Libre, and I think the thing that annoyed me about that is it wasn't always terribly accurate and it didn't allow you to make any kind of calibrations. So, when I moved on to the Dexcom I found that better, but then the Dexcom app on your phone is horrendously annoying in terms of like constantly alerting you if you're high or low. And it's like, well, I've already treated it that and you can't, it's really hard to turn the arms off. So, now I just don't have the app on my phone, I just literally just use my pump and that's enough. Like, I find it really handy just doing that. Yeah, that's about it.

Interviewer

Yeah, definitely. I had exactly the same problem last night. I was telling some of the others. So yeah, I'm a little sleep deprived as a result of those painful alarms, but.

P02

Yeah, it's just, it's nuts. Like. I mean, that's why I just don't have on bother having it on my phone anymore. Because if I'm going low the the pump yells at me so and that wakes me up so I don't need like constant alarm every 2 minutes like on my phone that you can't make quiet so yeah I agree.

Interviewer

Yeah. Do you mind taking someone else to?

P02

Oh, sorry, [P16].

P16

Yeah. So, I'm on the Medtronic 780 pump with the Guardian 4 sensors. Um. And I've been on that for about a year. I've got the Fitbit Luxe. And I don't, there's nothing I particularly don't like in terms of, like diabetes technology, but I think just when the sensors don't work, it can be really annoying. Um. And then also the two hour warm up period can be a bit frustrating sometimes. But yeah.

Interviewer

Yeah, no, I definitely agree with you there. Totally. Yeah, [P10], have you?

P10

Oh yeah, sorry. It was finding on my phone unmute. Yeah, I've got the t:slim and Dexcom G6 as well. So I'm quite lucky with that because I was, I managed to get it because I'm still in paediatricians because they haven't made me on. Because I got the pump literally like a week before I was meant to move so then I got like a nine month grace period. So, I was able to get it under the kid guidelines which was slightly easier. So, I was lucky with that. I've got the Versa 4 for the Fitbit. And then in terms of tech, I don't like I would kind of agree with [P16], I really, the two hour warm up periods I get, why they're needed. But like literally just I think two days ago I had the two hour warm up period and then a pump insertion and then the pump insertion didn't work because it like went against the skin. So of course, I'm getting no insulin for two hours and then when it clicked on, I was about to go to sleep and it was massively high. So then had two hours sorting that out. But no, that's the only thing. Otherwise, it's like the Dexcom’s been quite good.

Interviewer

Yeah. No, that's good to hear that otherwise that hasn't been too bad, but yeah, definitely those warm up periods are a pain. Yeah. And you kind of see there's quite a range of different technologies people are using and wearables as well. So yeah, it's quite good to get you all here together. Kind of the way this run is kind of I'll ask questions generally such just chime in if you've got points on it and you know follow on from people. It'd be great just to hear anything you've got. First off, I guess just kind of reviewing the past month again and thinking about the devices, what feature do you think you've used most most consistently or commonly?

P05

I keep quite a tight eye on like the zone minutes. Just cause I've been making like more of an effort to be active and basically I've just come back from holidays and I basically just being like a slug for a week. So, I was like, I really need to do something to sort this out. So, I've been using that a lot like when I have been exercising just to see what I've actually been doing, like with the heart rate and zones and stuff. And then I really like the fact that it gives you a readiness score as well, because I don't sleep particularly well of late. So, I've been like keeping quite a close eye on what my actual sleep has been like during the night. And you know, it's easy whenever you're like "ohh, I was waking up during the night", but whenever you're able to see numerically, like, "oh, you spent this amount of time awake, you were in this light sleep for a bit, you know, you were in deep sleep", whatever. So, I like the way the readiness score sort of brings it all together and it's like, "OK, so you've had quite a lot of activity, your sleep hasn't been great, so maybe take it easy today". Which you know, your body tends to be able to tell you anyway, but sometimes it's just reassuring when you have the wearable with all the data that it's got about you being able to say you know it's OK to kinda have a rest day.

Interviewer

No, definitely is that anyone? Is that something anyone else has use, kind of the zone minutes or readiness score?

P01

I I also find the sleep score thing quite interesting. I think that is like useful. I wouldn't say it particularly like makes me change any like behaviour or anything, but it is like interesting to have data on like how long it takes me to go to sleep and how much I wake up. It's made me more aware of like I need to have a better sleep. Ohh nearly spilt my coffee. I need to have a better sleep schedule. Like it explains why I'm so tired all the time.

P05

Ohh yeah 100% like cause one of my things has been like you know when it gives you like your month previous or like and it like there's like the little badges that it gives you. And it's like "this week you slept like a dolphin" or whatever.

P01

I don't think mine has that one, but it just it just gives a score, like a rating, yeah.

P05

Yeah. So mine is like “ohh, you know, this week though this past month, you've slept like a dolphin, which means that you've been averaging between like, six and eight hours, which is like typical of most mammals” or whatever. But one thing that I've found as well is like, it marks your. Well on my one anyway, it like tracks your sleep start time and like wake up time. So that's one thing. Yeah. Like, that's one thing I've been, like, more focused on is like trying to have, like, a set bedtime just to try and improve sleep overall. Um. And also just like a better routine in general.

P01

Yeah. And it's it's just interesting how much of the time that I I think I'm asleep for that it says that I'm awake for. And it does like, I think it's pretty accurate, it ties in with like, if I've had to wake up hypo or something. Or if a noise is woken me up so.

P05

Yeah, definitely cause I've had a couple of times where I've woken up for a hypo and maybe been awake for like 25 minutes or half an hour. Like obviously monitored and making sure my blood sugars have come back up and making sure that I've not ended up like rebound high. And then, yeah, maybe it's taking me like the rest of the hour to get back to sleep, but then the Fitbit has been like, “ohh, you slept from 10:30 to 3:30”, and that's all it's tracked. And I'm like, “no, I did definitely go back to sleep”, but it almost like tracks it as like a separate sleep instance, as like a nap. For like a few hours then.

P01

Ohh, it's like thinks you've woken up for the day. Yeah, OK.

P05

Yeah. So then it's like your sleep's being terrible. And I'm like, “not necessarily accurate, but OK”.

Interviewer

Oh, [P10], did you? You unmuted, was that something you wanted to say?

P10

Oh no, it's just in case. But no, I I have been using the sleep one as well. I think it's kind of same as you, not so much like change habits and like, validate how tired I am that day. I can kind of see that and it makes me a bit more kind of reassured. Because I always had, I did my A levels, and I had it during that and it was quite interesting to see sometimes the lack of sleep. But, it was nice to kind of like, reassure, it's not just me feeling tired, like there's a valid reason. So, I think it's more kind of the other way around that I use it for. It's quite nice.

Interviewer

Yeah, I think that's quite interesting. Has anyone actually made changes based on what they've seen or is it just more drawing those links?

P05

I'd love to say yes, but no. And I know a big part of that is because, like I'm on summer holidays at the moment, so I literally just have no routine because I'm not at work. So, I'm just like every day's a weekend. Woo. I will plan at some point to like, tighten up my sleep schedule to get ready to go back to work, because otherwise I'll just be like a zombie until Christmas.

Interviewer

Right. That doesn't make a lot of sense. Umm. And how about you, [P16]? Have you ever kind of changed anything as result of something you've seen on the watch?

P16

Um. Not really, I don't think. But if I know that, so because of the steps on it, if I know that I've done a lot of steps, then I'll probably manage my blood glucose a bit closer. Just because I know that it will then affect me probably for a few hours after, and possibly even the next day if I've done a lot more steps than usual.

Interviewer

Right. Yeah, that's really interesting to him. I mean, does anyone else make changes to their diabetes management, as a result of what they've seen on the watch?

P10

It's kind of like, the one thing I'm more aware of is like when you're walking from, like, about, is how much like activity that is. Before you kind of just think, it's like, liminal space, you don't really take into account of it. But because it is actually, like buzzing on my wrist, saying you're exercising, and you've been doing it for like, 40 minutes or something. I'm thinking I'm not doing that many changes, but I'm kind of more aware. Yeah, because I normally have. I used to always have a snack before I went out anyway, so I haven't had to make changes, but like, I'm kind of more, "OK, I'm doing this

specifically for this” and I kind of see how long it is, whereas I wasn't really taking that into account before.

Interviewer

Right, I see. UM, I guess, what about you, [P02]? You’re using an Apple Watch, so a bit different. Like, how does that compare to kind of this stuff?

P02

Yeah, sorry. I think the recording just kicked me out of the. I don't know why. So, if I disappeared, yeah. What was it? What were we talking about, have I?

Interviewer

Well, just I I guess in the last month, what have you been using? Have you kind of made it or do you ever make changes the result of things you see on the watch or might be related to type one diabetes?

P02

To be honest, no, I haven't used any of the data from the watch. Sorry, I've realised I forgot to mention I have an Apple Watch. But, I haven't made any changes to my diabetes care from the watch, but I think I have found generally that um I'm wearing the watch more. Erm. And to be fair, I've been paying more attention to the activity rings lately, mainly. Partly because my job has been like I've gone through a lot of different changes at the minute with work and things, so I just recently started, became a carer doing care work for a bit. So, I've been wearing the watch a lot more with that and that's handy. Like this sounds really stupid, but just wearing a watch helps me be a bit more time to places. But, also the activity rings is really more interesting to me because sometimes I feel like, you know, I feel tired, but you don't necessarily appreciate how much activity you're doing until I get home and see that I've closed all the rings. So just from yeah being out and about. So yeah,. No direct sort of changes to my to my. From the watch to my diabetes care. But I have sort of, being part of the study I guess, has kind of made me a bit more aware that of sort of what my daily CGM traces are looking like. So, I have. Yeah, like the new job that I sometimes have, like breaks. I start very early, have a break in the middle of the day and then go work from, till sort of quite late at night. So, I sometimes would have a nap and then another day, so I had that was wreaking havoc with. I'd eaten and go for a nap, and that just wasn't great, so had to kind of was looking more at my data to try and tweak things to try and stop that from happening, but. Yeah. So just I think. I suppose to sum up, increase general awareness of what's going on.

Interviewer

Yeah, I think there's a couple of really interesting things I'd like to talk more into. First of all, is just kind of you commenting on that change. You think that you've been looking at it a bit more. I was

interested to hear of what other people are finding. Do they think they're using it the same amount they were the previous month or more or less so?

P05

I think I'm probably.

P01

I think. So sorry.

P05

I'm using it quite. It's like, oh, sorry [P01]. I'd say I'm probably using it slightly more. But again, just because of that increase in being active. But I still think that I'm very much in the novelty bubble with Fitbit because I'm like “ohh, what can I do with it?” And like like, I'll set an alarm if I've got like the tumble dryer on or like I'll put a timer on for if I'm making food or something. So, like I'm using it generally like for a lot of handy things that maybe previously, I would just have used my phone for, but obviously because it's on your wrist you can just like bam to it and it's there and you're not like trying to listen out for an alarm because it's buzzing on your wrist, so it's fine. But, I definitely say that like my like usage level has been pretty constant, but probably like on an upward trend the further end of the study that I've gotten so far. Sorry [P01], do you want to go?

P01

Ohh no sorry, there was like a lag in it so I didn't mean to interrupt you there. But yeah, I think I also find the alarms really useful, which is really silly because that's such a like basic feature of a watch. But I've just never thought to wear a watch before, I don't know why. But it is so much easier, especially like at work, not having to get my phone out and stuff. Yeah. So that that is useful. I think I probably use it like a similar amount to last month. But maybe slightly less in terms of like last mont, you know you can like log you can categorise like the exercise that you do so it like automatically picks it up and you say what you were doing. I think I was doing that like every single time last month and now I've kind of stopped doing that. I guess cause of the novelty wearing off slightly, but I still look at the actual data, the same amount, I'd say like in terms of the sleep and the activity ratings.

P10

Yeah, I would say I'm I'm quite similar like I've used. I think I've used it slightly less, but I'm using certain features more. So, like the exercise one partly because I'm being more active this month but like I'm I'm still like tracking each one, but I'm just relying on the automatic one. But, then I'm checking the like the data for my runs more, but I'm not looking at at my sleep as much. So, overall, like I kind of changed how I've used it slightly as I've kind of gotten used to more functions. And like there's sleep, I'm not really checking nightly data, but I found out you could like you can do the

average like what time you want to sleep and what time you woke up and I found that more useful. So, I think I've kept it roughly the same, how, the amount I'm using it but I've changed slightly how.

Interviewer

Yeah, no. Yeah. I mean, it's interesting to hear there's quite a range of stuff in, in terms of how that's changing over time and the different uses. Maybe maybe going back to another point though a couple of people were making was just around kind of changes people have noticed in the last month. I think like [P05], you said you were on holiday and [P02], you mentioned those kind of changes in what you were doing. Has that been kind of reflected on information you've seen on the watch or is that just kind of purely you, you know, that's a change you've had?

P05

Yeah, like I know. Like one of the days that I was on holiday, obviously like hadn't done very much was like maybe averaging like 5000 or 6000 steps a day, so like a lot less than like what I would normally have when I was like in a normal routine. Um. And I used the swim function on the watch and was like in the pool and like was in there for about 25 minutes, just sort of like doing some length. And then ended up having one of those hypos that just kind of creeps up on you, but I knew something wasn't 100% right. And actually I'd used the sensor and I was like, that doesn't seem reflective of how I'm feeling, so actually did like just a an actual blood sugar. And my bloods are 1.7, which was really scary because I felt perfectly lucid. And I literally just sat down and said I, my daddy was on the phone next to me to my uncle, and I just showed him the blood monitor and just went “help me”. And he was like, “Oh my God”, cause those are just, like, not numbers that I'm ever used to seeing. And that like it it really scared me because I was like, “oh my God”, like I have had blood sugars of like maybe 3.1 and I have felt like I'm about to cry. Like I just felt so out of it and like, really lacking concentration and really lacking focus and you know. Whereas, that was a completely different sort of hypo experience for me and I recovered quite well, I was like, fine within half an hour, and like a little bit shaky. But, I don't know if yous ever get this, but sometimes like if I have a like a hypo for maybe longer than anticipated, or like I've woken up with one like, I feel like I can't feel my face. I get like pins and needles in like my gums and like my face. And I'm like “ohh”. And it's a really disorientating feeling. But I could directly correlate what had happened from how I started going for a swim to ending up at that stage where the blood sugar. And you know, I was like sitting reading my book, it was like 35 degrees of heat, and got into the pool, started being quite active. Had checked my blood sugars before and they were like 6.4 and I was like, “perfect, we're fine”. And I had actually taken my pump off when I was swimming. Purely because, like the Medtronic is fully waterproof, but mine has like a small crack in the battery casing so just best not to mess with that. Um. So, I had my pump disconnected as well, so obviously for the like 25 minutes I wasn't getting insulin and it still happened. But obviously that increased activity, you know it it does make sense. But, I guess that was a point where I could see the direct correlation between what I'd done, like the recording of the activity, like with the swim function. And then obviously it gives you the details on the app of like this is how far you went, this is how many lengths you did, whatever. And then the drop off a cliff with the blood sugars. But I suppose that was like the big kind of example that I could think of. From the past month of usage anyway with regards to like seeing that correlation between the activity and then the blood sugars.

Interviewer

Right. Yeah. Has anyone else notice like a similar point of correlation with that?

P01

I had a a similar experience actually at like exercise class where I just suddenly had a really bad hypo and it was like. I don't often, I have quite a lot of hypos but they're like I'm usually sort of like, they're manageable ones. But, it was like a a step class and it was like one of those ones where the floor like feels like it's like going like that and like your vision starts to go. And so, it was really weird cause like everyone's like moving around you and it felt like I was in slow motion and everyone else was moving really fast still. And like, because everyone was moving, it was really hard to like, walk to get my, like, dextrose because, like, the floor feels like it's moving, like people are moving and I'm like, “whoa”. But so that was quite a like sudden and bad one and I guess yeah, it correlates with the like Fitbit data because it was in the middle of like an aerobic class thing I guess. But, I wonder whether it's also like got a lot to do with the activity like in the rest of the day, because I like I go to that class every week, so it's not out of my routine and I'm normally fine. But I think a combination of like having quite an active day and like not sleeping well and being stressed like generally I think. I don't know like what part of it, but like I feel like it's a combination of things that leads to ones like that. And also similar to what you were saying, [P05], about like feeling different at different like blood sugars. I often find that I'll have a hypo where I'm like, really, really low on the reading and I feel fine. And then sometimes I have one where I'm 3 point something and I can't stand up. It's. I don't understand what it is, but the numbers don't correlate how I feel.

P05

I know. Like have you ever had one where you feel completely fine and then all of a sudden it's almost like you're going blind?

P01

That well, that was this one in the class because I checked my blood sugar at the start. I think I'd had a really high day and I'd come down to like 10 or something before the class started. And I thought “that's fine, it will like, that'll be fine for the class”, but it just was like one of those times where it's been really high and then it like crashes. And yeah, it was, it was like really sudden and like vision starting to go and stuff.

P05

Yeah. Which is then really scary, especially when you're in like a public place, like at a gym class where maybe you don't really know people and they kind of are looking at you funny and you're like, “no, I don't actually want this to happen. This is like somewhat beyond my control, unfortunately.” And it's just it's a really. I think it puts you in a very vulnerable position I think at times cause a lot of the time, people don't understand conditions, especially with like diabetes and.

P01

It's such a complicated thing to like explain to people. I think even like I had to have, like, a surgery on my teeth recently and even just explaining it to the nurses in the hospital there, they were just nurses, they weren't diabetes nurses, and they did not understand it at all. They were, so confused. They were like “what, if you're low, do you put do you use the pump or like”. They didn't, no one understands when you use.

P05

People don't get it at all.

P01

No and then the fact that you have to eat something sugary, which sometimes is sweets or looks like sweets when you're low, is such a like joke sort of medication. Because you're, like in this gym class I'm sat there like feeling like I'm about to die in front of everybody in this class including, it's one at my work, so including like my boss at work and stuff, and they're all carrying on. And I'm sat there, like at the front, sweating and and like, yeah, just looking completely out of it. And then I have to eat like sweets in front of everyone, it just feels so, like, silly.

P05

Yeah, it is. It's a really weird thing because like one thing that I get a lot is people are like, “so you can't have sugar at all?” And I'm like, “no, it's not an allergy”. Generally speaking, no human being should be like drinking 2 litres of full fat coke and having like a Mars bar for lunch every day like, that's not healthy in general. Technically, if I wanted to do that, I would just have to bolus correctly for my meals. But it's not recommended that anyone do that, particularly if you're type one diabetic. And then they're like, “so when do you need sugar” and “like, if you collapse, would I just like stab you with your pen?” And I'm like “firstly, I don't use pens anymore and secondly, no, that would kill me”. Like people just don't get it at all. And you're like trying to explain it's like potentially lifesaving information to people and they just look at you like.

P02

One of my pet, one of my pet peeves, people who are like “ohh yeah, no” or they'll say. Oh, you know, I'll, I'll say to them, “oh, I have type one diabetes” and then they'll be like, “oh yeah, my granddad or someone has type 2” and I'm like, “yeah, not the same”.

P01

That’s not the same.

P05

That's not, literally not the same.

P02

Yeah, please, thank you, but it's not the same.

P01

Or you don’t look like you should have it?

P02

And then I'm like, “well, no, no, it's both”. And then you try and explain that it's both like and they'll still look at you confused and it's kind of like. Some people will actually listen to you and they'll get it like when you sit and explain it. But it's when people like you tell them something cause you think they're, you know, they're interested and then you'll start to explain it and then you can tell they're not really interested, they've stopped listening, so then they don't they don't bother to internalise what you told them. And then you're just sitting there, like, “well, what was the point in me trying to explain it then?” You know what I mean? That that's one of my pet peeves I have to say.

Interviewer

Yeah, I I feel like it is and diabetes, well, type one diabetes can be such a complicated condition to explain even the basics behind. And then you're kind of adding to that with quite complex technologies, you’re now beginning to see kind of used or as part of it. I mean I how do you all feel about that? Do you want to have that understanding of exactly what the technology's doing before you use it or are you happy just to kind of trust it and just go with it?

P02

In in terms of like CGMs or verging into like new tech?

Interviewer

Anything. I mean, and I guess, yeah, CGMs when they first came out, but people, some people were quite worried and there are still people who don't want to move across to it. But, then also the closed-loops more recently and potential future ones.

P02

I think personally, I am all for like new technology as long as there's a way like you can manually adjust it if necessary. So, like I just mentioned earlier, like you know when Libre first came out, it was amazing like it really helped, but then you know and I've self-funded for a while, this is like you know,

yes. But like it's pitfalls, downfall is like that you can't calibrate it. Like you know, because like we're saying like, there are times when you have a low and even with the Dexcom it's you know you have these moments where given that, it's like, you know, interstitial fluid, it's never going to be exactly the same or it's like on wherever it is on your body. Like, there's going to be times when it doesn't measure up to how you feel, and on those times, you might really think or like times I've gone away and done a blood sugar and gone "yeah, my sensor's gone a little bit out, I want to calibrate it, so it's not going to tell me I'm high or low for the rest of the day". Um. And I feel confident that cause I can do that with the Dexcom. Like I'm happy like cause 99% of the time it's accurate, on those 1% of the times where I need to go in and make a change that I can. With the Libre, because the, at least the early versions you couldn't do that, it it was kind of like, "can I trust it?" And when you can't go in and and then you sort of think "ohh I'm not happy about this" and take the sensor off and then you've wasted a sensor and that kind of thing. Whereas yeah, I feel as long as there's as long as you know, we have a general understanding of how it works and there's a way for human to go in and like manually check and adjust it if needed, then I'm kind of happy to like yeah, I kinda hope that technology keeps getting better and smarter um to help us.

P05

Yeah, I think I like, I've very much agree with you, [P02]. Like I personally, I'm like really excited for the new um guidelines to come out about the CGM. And obviously like nothing's confirmed yet because it hasn't actually been released, but I know that my DSN and the pump consultant that I work with in [UK city] are like really hopeful that it is going to be a positive change and that it is going to become available, you know. And like the team that I work with in in [city], like they have been so supportive and they've been amazing. So, they were like "if we can get you on it, we are gonna do everything that we can to get you on it". And I was like, "oh my God, that would be amazing". Like the thought of being on a closed-loop system and particularly obviously using the AI. Like even now, like looking back, it blows my mind how much the world of diabetes and diabetes technology has changed since I was diagnosed like 16 years ago. And like even little things like it takes 5 seconds for a blood glucose result now. I remember having strips when I first was diagnosed and it was 15 seconds for a blood sugar reading and it was like. And do you remember like calibration strips and all that kind of stuff? And you know, that was minor at the time, but now looking back and it's like, "oh, we don't even have to do that anymore". And we've got pumps and you know, even like having FIASP, like the new um, like faster working insulin and stuff like that, like the changes have, I think have been really positive. But picking up on what [P02] said, like I think you know, it is still technology and technology can fail. Um. No matter how big or how small that might be, or how life savingly important or you know anything from your Casio wrist watch could break to your lifesaving diabetes pump. Um. So, I think definitely having the ability to intervene and always being able to go back and and readjust and make sure that it is working for you in the way that you needed to. Because diabetes is so changeable, um you know? And I always like say to my parents, like I could do the same thing like two days in a row. I could have the exact same amount of water, I could have the exact same amount of carbs, I could take the exact same amount of insulin and I could almost guarantee you that my blood sugars wouldn't be the same. Like they might be similar, but they wouldn't be 100% the same, like it's not that exact. Like the science has gotten more exact because we have, I suppose better and more precise tools now to manage it and to intervene when we do need to. But I think I am all for the progression of the technology, provided that there is always going to be a failsafe that I can turn off the AI and go back to, you know, background basal that I have preset. And obviously I know that the Medtronic pumps do have that function. I was speaking to one of the representatives at like the

show and tell day and they were like “ohh, you know, like we would recommend that you know, if you're really unwell for like a day or two, or maybe if you're on antibiotics and like, maybe that's making you worse or like it's affecting your blood sugars or things like if you're on steroids for a short space of time, we would always recommend turning off the AI function because we don't want the pump to learn those patterns of behaviour”. Because obviously that's not your normal day. That's not your normal way of working. It's not your normal bodily function, so to speak, so. I think it's good that the sort of teams that are behind this are also aware of that advice and able to give that back to the users. Because as a, you know, a service user, which ultimately I guess we are like. It is still a business I suppose. You know, because it's healthcare, they still have to follow that duty of care and make sure that they are doing what's best for us to some degree. So, it was it was reassuring hearing that from like the representatives from Medtronic being like “oh yeah, like we've already thought of this, and you know this is how we're going to circumvent that or how we're going to combat it”. And I was like, “ah, OK”. So, like pretty much everything that I had to question, they had a well-founded answer for. So, it definitely gave me more confidence going forward and like I suppose, quite a hopeful disposition for the future, I suppose.

Interviewer

Yeah, that's that's good to hear. I guess it might be interesting to hear from you, [P16], as someone who's moved on to a closed-loop system, how you've kind of found that process and trusting it.

P16

Yeah. So, I think moving on to it, it was quite scary because I was like, “oh, my goodness, I've now gotta put all of my trust in this and I don't get to pick how much background and then I'm getting or how much correction I get”. But, I think as time has gone on, I've learned to trust it a lot more. But even now I think it is still quite scary seeing it correct you and seeing it do things that you're not in control of. Because I think I've spent so so I've been diagnosed for 10 years, so I spent so long being in control of it myself that then putting my trust into the pump to do it is quite scary. Um. And also I've just been ill so I, but I did leave it in auto mode I didn't take it out because I didn't want to have to deal with it and trying to sort my level out myself, I didn't want to have to deal with it. But the pump didn't cope that well. But yeah, it is good when it works, but when it doesn't, it's quite difficult.

Interviewer

Yeah. Do you think you were given kind of an adequate enough description of how it worked for you, to begin with?

P16

Probably not, because I don't. My hospital team aren't very good at all, in the fact that I have, like, barely any contact with them and. It was kind of just ohh, time to change pump, here’s the new one. But then obviously I did me and my family did research ourselves on it to kind of see what it did, but yeah.

P02

Which pump were you on again? Sorry.

P16

I'm I'm on the Medtronic 780.

P02

Is that one of the new like newest ones?

P16

Yeah. So it's it works with the Guardian 4 sensor and it's the closed-loop system.

P02

OK. No, that's that's just really interesting because like just going off what you were saying about it, not always working that well. Like at least being on the t:slim with the Dexcom, it's. Like there are times when, I find that it works reasonably well when I'm fairly active, like you know, but if I'm quite sedentary. For example, if I'm having a day off, I'm like on a lazy day I think, I think the way the the t:slim works is that you will correct based off of basal profile that you put in, but then it will it the algorithm will like kind of use what you what you've put in as the base but then try and correct off that. And sometimes, I think it doesn't always work that well when you're not, when I'm not doing very much sometimes because. And even you know like for example cause I wear my Dexcom, you're probably not supposed to, but I wear it on my leg because it's comfy as close for me. Um. And I sometimes think that because you're just not moving that much, the exchange between blood and fluids is not happening as well. So I think sometimes glucose will like accumulate in in like in the fluids rather than be in the blood and so the pump thinks I'm really high and I'm not really. I feel like it's like, yeah, there's times when I'm like “God sake, like you know, I'm not actually, you know, 20 or whatever, I'm fine”. But you just have to kind of kind of, I don't know, turn half a blind eye, I’m fine. But yeah, I know what you mean.

P05

[P02], would you say that you're, like, quite confident in your system at preventing hypos then, like whenever you said, like, you're when you're at work and you're quite active, it works at, like, sort of reducing your your insulin or whatever to then.

P02

Yeah, so I, my favourite favourite thing about the t:slim is the night mode because it just stopped me going. I used to get lows a lot in the night before I went on, and my major problem was I'd be like

when I go into clinic like look at my thing, they're like, “you know, you're not actually that bad”. But I'm seeing, they're seeing. The problem was I used to change when I do temporary basals overnight because I'd be trying to prevent myself from going low. And then obviously that those temporary basils would have human error in, that I'd wake up and 90% of the time I'd be too high. So, then I'd have a terrible day, because I'd be too high, over-correct, go low, go high, go low, that kind of thing. So I like, really, really trust the pump on night mode because it just like, saves me like I can go to bed and and it would it literally just sort it out overnight. And like I'd say I very rarely have lows overnight now because it works. Um. I have to try, like there's more of my intervention during the day for low prevention at the minute. Like as I say, I'm just going back to what [Interviewer] was mentioning earlier about the activity correlating with that, the watch data and stuff. You can see I've I've like my work at the minute is new to me, so you know, I do keep a very close eye on what I'm doing because I don't want to be low, you know, with a client who I don't know, like in their house. Like, who is you know? So, I keep very close eye and I sometimes have like preventative hypo snacks, I use Jelly babies I like yeah, if I go in. But I also if I know I'm in a, in a rush I'll also put on exercise mode, which kind of sets you like at a higher level, so it will suspend insulin at a higher threshold so you don't drop. But yeah, there are. Yeah, I’d say night mode. Yeah. Night mode is my absolute favourite feature. Um. The daytime is still good, but like you need, like I find if my lifestyle or like has changed substantially or like the core basal profile no longer is accurate enough to reflect what I'm doing, I have to go in to the core bit and change it because it always will work off the the kind of the baseline, if that makes sense. So, if my baseline is way off, then the corrections or the the system will be way off in trying to make automatic ones. So that's that's what I found.

P10

Yeah, no, I'm quite similar because I'm on t:slim as well and I found yeah, because I had that issue with the baseline being off. And because it was during exams I was doing no exercise at all whatsoever and then like coming off into the summer doing quite a bit and like it just didn't work. So, I don't know whether you, you probably do, but you can do the different profiles and like I found that I switch even like during when I was at school I like had a school day one and a weekend one and that helped a bit. But yeah, I would agree though, the sleep one is. I used to get night lows all the time and I haven't had one about since I got it. I don't think unless or.

P02

Literally, yeah. It's so good, isn't it? Yeah, but no, I'm the same. I have different profiles and like you know, if I'm being a real lazy toad. Like then I’ve got a really extreme profile, but it's quite I. A weird thing about me, I'm gonna reveal myself, I label all of my basal profiles with the weird names, so when I go to the clinic, my clinicians look at me and “why is it called like some weird name?” But I’m just like you know what, you gotta keep things amusing.

P05

That's not that weird. I've named all my pumps, so.

P02

Yeah, but. Yeah, it was. Well using one called Stan at the moment or something. But there we go, stop being weird.

P05

Yeah, well, my pump's called Frank, so. The first one was Jerry.

P02

Love it. Brilliant.

Interviewer

Yeah, yeah, I guess people have mentioned a couple of times, there are still those events where things go wrong. And that, that's probably something that happens with technology as well, as it becomes more complex. You know it can become more difficult to manage it. I mean, I just had the example there, I was trying to put in a, test my blood sugars and tried to put it into the pump just to get a correction and put in to have it suddenly was saying "ohh we're about to deliver 11.9 units" rather than. So, I put it into like the deliver point rather than the blood glucose one. So, there is kind of those options. I mean, you can cancel obviously quickly, but that, that panic, how do people kind of react in these instances where things go wrong?

P10

To be fair, with my pump suggest I always. So, like with the t:slim you can do like the amount of carbs and it tells you how many units. I never trust, like I don't not trust it, but I always think I know better. So, I use it as like a starting off point, then I'll randomly add a few units or minus a few units just based on how I'm feeling. Like my, whenever I tell my nurses that they always look me in horror, but yeah, no, I just kind of play around with numbers as I feel like it. But it seems to work so like. And with my the pump. So I do that a lot with like I'm putting in 20 grammes of carbs, but I've actually just put in 20 units but I think the t:slim is quite good, if it's an unusual amount, you can kind of set a limit. So, if it's over like, I never really have units over 10 units at a time, so I have that limit. So, if I accidentally type it in, it won't deliver it. You have to like manually to do a password and stuff. Not password, you have to tap those things on the screen to let you deliver. So, I think the fail safes are quite good on the t:slim with things like that. But, obviously, you can kind of still do the extra 5 units when you meant to do like .5 or something, in which case I just down some Lucozade but yeah.

P01

Yeah, my Medtronic one has a similar thing where I think I've I've got my limit as like 10 units as well. And so if you wanna go over that, you have to do it in like 2 separate boluses, so it doesn't make you sort of think about it. But I also do the thing where I'll randomly add on units, but it tends to be like when I'm high and if I've been high for ages, I'll do like a rage bolus, but then it it is never good. It

always, I always then have a crashing hypo later, so I probably should trust the technology more, but that's just my impatience when I have been high for ages.

P05

Yeah, definitely. Because it's so frustrating. Like I'm. My consultant, always says like “ohh if you're high, like if you can try and sit in your hands for like 2 hours and like just let the insulin work”. And I'm like, “but it's not working quick enough and I feel like crap so”.

P01

Yeah, that's the thing, when you feel awful and it's a busy day and you've got to, you've got to do stuff like I don't, it's it's really hard. If you took like time out every time you felt ill because of your diabetes and literally.

P05

You never get anything done.

P01

No. I literally wouldn't like be able to work. I wouldn't be able to do anything so.

P05

Exactly.

P01

The amount of time you just have to like get on with it, yeah.

P05

Oh, I know. And people just don't get that either, because it's like. Like what you were saying, like rage bolusing and like, I find I get really crabby when I have a high blood sugar. Because I feel a bit crap, my head might be a bit sore, I'm not feeling 100%. Like you just don't feel good. And when that sort of prolongs and then it's going on in the background and I'm checking blood and I'm like increased basal rate and I'm like, you know exactly the same thing. So maybe my blood sugars have been sitting at like 14.5 for like over an hour and I've already given a bolus. So, then I'll put it back into the Medtronic and it'll be like, “oh, you already have this act of insulin, which is doing its job” allegedly and not known difference has been made. And then it's like “we recommend 0.25 of a unit”. And I'm like “no, another unit and a half”, smash it on there. Let's just see how it goes and then

we'll deal with the consequences when my bloods inevitably fall in like 2.5 hours off a cliff, and then it's like trend arrow straight down and it's like 9.8, 7.8, 6.5 and I'm like “there it is”.

P01

I also find with with the Libre that like because it does the alarms for like being high and stuff. And like other people that know me, know like what the alarms mean and it alerts like everyone else to I guess how how bad my control is sometimes. So like in, in some cases it's it's useful because they can then have that understanding of why I'm in such a bad mood, but also I I feel like sometimes people like very well-meaning but they think that it's helpful to be like “it’s going off, you need to do something. Go on. Treat it.” And I'm like, “no, you don't understand it, like I have treated it it's there's nothing I can do now, like I just have to wait”. And it's almost like there, it feels like they're frustrated like yeah, the fact that you're doing nothing and the phones like screaming at you. But so, it kind of does encourage other people to voice their opinions on that more, which can be a good thing, but is often, often just frustrating.

P05

Yeah, 100% because like I was diagnosed when I was 10 and like really quickly, like took on my own care. Because, like, I didn't want people poking with my needles. I was like, “I very much want to do this myself, please, you know, I would like a safe working space where no one's touching me with a needle”. So, I like really much like took on my pens and stuff on my own. But, then obviously like as I got older and like moved out of the house, when I went to uni and stuff like. Because I wasn't living with my parents and my family, they naturally became less involved in my care. So, even now sometimes I'll find that like if I randomly have low blood sugar, but maybe it's because I gotten up and like went to walk in the morning and then have come home and I've been like hovering around like 5.5 and then suddenly they'll drop and we're meant to go and do something, my parents will be like “why are you low”. And I’m like “firstly, literally like it's not my fault. Secondly, do you think I want this, because no, no one has ever wanted this ever”. Or like if I ever like, there's a lady that sits next to me at work and one of our mutual friends is type 3 diabetic. So, she actually had like pancreatic cancer and ended up having her pancreas removed. So then now as a result of that has diabetes, which is like. She's in like, a minor, like percent of people who have that condition. Well, through those circumstances. So, like, she knows a little bit about diabetes, but not a lot. So, like, there's been times where I've, like, maybe really overcorrected a hypo because I'm about to go on and go start teaching for the beginning of the day and then it's break time. And because it's like on the Libre particularly like if it's green, it's like all beautiful and like lovely and nice green, smooth colour. And then you get like.

P01

It's very visual, isn't it? For other people to see the colours.

P05

Very visual. Ohh and then you get.

P01

They think it's as simple as green or red.

P05

100% and then you got like orange or red and everyone's like “bad, bad”.

P01

Bad diabetic, yeah.

P05

And I'm like, “oh, just leave me alone. First, I didn't ask for your opinion. And secondly, it's not that simple” like. Yeah, I've taken maybe one too many glucose tabs, but it was necessary because I can't be standing in front of a group of year 8s in a midst of a bad hypo because they're all feral like. Yeah, you know what I mean? And like I think because of that visual aspect a of it people then feel need to, exactly what you said [P01], offer unsolicited opinions. And it's just like I do not need this from you today.

P01

When when you're high or low and you're feeling crap anyway, the last thing that you want is somebody who doesn't understand it to, like, be asking you questions and giving you advice. Because it's like, “look, I already feel terrible can you just leave me alone to deal with it? I don't want to talk to anyone.” And it's interesting what you said about your parents as well, because I'm like similar situation, I've been diagnosed for about the same amount of time and obviously I live like on my own now. And recently I went like back. I went on like a day out with my mum and I had like, quite a low hypo. But it was what it was, one where I was still, like felt OK, and felt in control of it. And her, like, anxiety around it was just, she it just really like pushed her over the edge, she could not cope with it. And she was like buying me, like cakes and juice and just like buying me loads of like random food, like trying to get me to eat. And we were like literally on our way to go for lunch. And she brought me a huge, like cake covered in caramel and a big smoothie and was like eat these and I'd already had sweets as well. So, I was fine, treated it and was on my way to lunch and it was like, “no, you don't understand, if I eat that now, I'm gonna be so high for the rest of the day”. She's trying to help but because her like knowledge of treating it was when I was a child. So, like it's very different controlling diabetes when your child to an adult and like a teenager as well.

P05

Oh yeah.

P01

So, she her like the last time she helped me with it was when I was a teenager and I was on like injections and fingerpicks. So, there was a lot more like, I I guess you had to think more about a blood sugar reading because you couldn't as easily just get 1-3 minutes later. And so, I I feel like yeah, she's maybe not, like because she's had been away from it, she's not up to date with like the technology. And you do forget how much it changes, how you deal with it and how you manage it.

P05

Yeah, definitely.

Interviewer

Yeah. It's so in terms of, yeah, I think it's quite interesting saying like this dosing in anger and sometimes it's triggered by alarms or other people around you. As a result of that, how do you feel about systems that maybe kind of limit what you can do or kind of trust themselves more versus systems that you still have always in your control exactly what's happening?

P10

To be fair, I quite like. So, I found my rage bolus, but I don't know the different pumps do it, but the extended bolus. It's the best type of rage bolusing. I would recommend because you can set it high and set it to, like I always do like 4 units in an hour and I never intend to have the full 4 units but for some reason because it's like a slow drip feed, it just gets me down really easily and it's slightly smoother. So, then I just stop the bolus, so I've only had like 1.3 units, but because it's been delivered constantly, it works. But I think that's my way of overriding the pump. Because the pump doesn't like it if I've given myself loads of insulin then it kind of stops giving me basal. So, I kind of with the extended bolus, give myself basal and loads of extra. Yeah, I think that's my way around it. So, I feel like the pump is obviously designed so you're not meant to be able to do like bad choices, because obviously they need to cover their backs for insurance and things like that. But I think it's quite good that they've like, whether it's actually or not, there is back doors, so it's just a matter of getting used to things. So, I think that is that kind of like toeing the line, where they don't encourage that kind of behaviour, but I still think you could be able to. Like it's your body, you should be able to like choose how like badly you want to treat it for want of better words.

P05

Yeah, definitely. And also, when I go to like clinic appointments and whatever, obviously they're all like Zoom based now, but you know. With stuff like that ultimately, I can't help but feel like I'm always going to be the expert in the room. Because a lot of times like people with, who are involved in diabetes care are not actually diabetic. Like the first interview that I had with [Interviewer], I was like, “are you one of us?” And you were like, “yeah, I'm also diabetic”. And I was like, “well, that makes more sense” and also like I kind of feel like it makes you trust someone more because it's like you've lived it, you live with it every day, you know what I'm talking about when I say rage bolus or like this or like post hypo I need a nap because I feel like I've just been on the brink of death. You know, stuff

like that. And ultimately, if I've maybe said, like [P10], what you were saying before, like whenever you say something and your nurses look at you in horror. But it's like but it works and it reduced the high, the high blood sugar. I didn't have a rebound high or like a hypo or whatever like. You know, so adding that extra unit and a half or reducing that unit and half was the right thing to do. And when you're in your own skin and you're like. Because sometimes you just you've got, like diabetic, spidey sense and it just kicks on and then you go “right, well, I'm gonna have to do something about this cause I can feel something coming”, whatever that might be, and then it happens and you're like, “oh, there you go so, I was right”.

P01

Yeah. I think sometimes like your body is going to be better than any technology. But I think in terms of technology, especially as I become more reliant on it I like sort of, I listen to my body less than I used to. I rely on the technology a lot more than I rely on my body now, whereas I'm pretty sure, like when I was on fingerpricks, I probably could have had good control and not done any blood sugar readings, just purely based on how I felt, whereas now that would be a disaster. But I think, in terms of like. I guess, like my my sort of mental health related to the diabetes as well. The less I have to think about it. So, like the more that the technology does, the better and I'm pretty like trusting of it as long as that is some kind of way that you could intervene if you had to, even if you never used it. If it was like some kind of system where like you literally couldn't change it at all, like it didn't even have any buttons on it you couldn't change or add anything I wouldn't, I wouldn't use that. I'd need to know that in an emergency I could change it. But, if it was something that the expectation was day-to-day, you didn't have to do stuff, then I would be totally on board with that. Just cause of like it would lessen the mental load of it.

Interviewer

Yeah, I was wondering, on this technology side as well. So yeah, if you see thinking now into the future of what diabetes technology might look like and there is kind of. I mean there is possibilities that it can factor in more and more things maybe from a wearable and kind of has that more idea, would you still want? You know and then it's it has a lot that it's kind of taken in to make its decision. Do you still want to know exactly how it's done that?

P01

I think understanding how it works and and why it's doing what it's doing gives you like that context behind it and I think that helps make, like, reassure me. And also it does mean that if something goes wrong, I sort of know how to deal with it a bit better. I think like the fact that when I was diagnosed, I started off on like injections and stuff has, means I have that like that sort of base understanding of how I would manage my diabetes if all of the technology failed me. I think if I was diagnosed now and was started straight on like the sort of best technology, there would be a danger of like if it went wrong, it just being really bad and sending me into like a panic and just not knowing what to do at all. So, I think like having some level of understanding of what is happening and why it's happening is always going to help you be reassured and make best use of whatever the technology is. But like obviously you don't need to understand it on the level that the people that made it do, just a basic understanding of it.

P05

Yeah, I I very much agree with what [P01] just said. Like I think I'm very much in the school of thought that, like knowledge is power. But, also with something as personal and fundamentally important to your well-being as managing a long term condition like type one, having that information available to you in a really manageable and accessible way is only going to make your, the benefits that you can reap from the technology better. But also, it gives you that safeguarding, safety net of should disaster strike and the pump fails or the AI goes crazy and tries to take over the world or whatever, like we can, you know, there's an off button, there's a failsafe. You know we can step in and you know, circumvent the worst possible outcome. But like just picking up on what you said [P01] about like kids who are like, just like, diagnosed now and just start on a pump. I kind of feel like, as amazing as that is, I feel like they sort of are missing like fundamental diabetes skills to some degree. Because like were like, did everyone start on pens?

P10

Yeah, I'm I'm quite interested because I was at diagnosed three years ago, so I was quite late.

P05

Oh yeah. Wow.

P10

I spent like a year on pens, a year on with just finger pricks. No, not even that, I was half a year on finger prick, it it must have literally like four months or something, then on to the Dexcom, a year with that then onto a closed-loop. So, I think it was quite an accelerated process, but I really completely if I didn't do the pens, I wouldn't have been good on the pump. And actually, what's interesting, my control is worse than the pump ever so slightly, but with the pens, because I had complete control, it took more, but I was able to get better control like I could go 4.0 all night perfectly flat line, but I was getting hypos and stuff. So overall management was quote on quote worse. But what's interesting about the tech, I'm just kind of like, use it as a safety net so I can choose not to use it all day and not have a mental burden. But I would agree, like if I didn't do the pens at the start, it would have been more reliant on the tech.

P01

Do you think your control being better on the pens was directly because of like the pens versus the pump, or do you think it was to do with sort of honeymoon period that people often get when they're firstly diagnosed. Because it's quite common, I think, to have, yeah, better control when you're first diagnosed.

P10

Yeah, I think I'm quite a weird case as well because I was on another trial. And they gave me a drug to try and extend the honeymoon period. It was really interesting actually. Um. But I was off that drug because it was just a phase two, one but. I think it was more. As a part of that exact trial, actually one of the nurses and the doctors, he literally just said to me within the first three months of diagnosis, he said, “like kind of ignore your clinicians”. He's just said, “go your instincts”. So from the get go, I was kind of like taking everything with a pinch of salt.

P05

Yeah, that's not necessarily the best advice for someone who's just had a life changing diagnosis who hasn't necessarily been given the information or the education to properly follow those instincts yet.

P10

Yeah, because it was mid COVID like I was not, I hadn't seen anyone apart from him. So, it wasn't like ignore them but it was like, you know that often. And I found with the pens I had more control, so I wouldn't even say it was honeymoon. I think it was the complete control, but it took so much time. I was like correcting a lot a lot and doing a lot of hypos and which is why I moved on to the pump and I prefer the pump, even though I'm slightly higher all the time because it's less effort. Um. So, I would say actually tech doesn't always make it better, but it makes it easier. So that's kind of my view on it.

P01

I guess it's more sustainable as well, to to be on the pump because you might have found after doing injections for like years and years that you got if if you're saying it was like a lot of effort, you might have sort of been a bit burnt out by it and then the control could have got worse. So, I wonder if the pump, although like day-to-day your control might not be as good, I wonder if over like a period of years it will balance out to be better?

P10

Well, no 100% I I had the luxury of like I was in school. So, though like I was doing homework and stuff like that, I didn't have that much on during the evenings and things like that. But during A levels with pens, it would have been a nightmare. The pump save me completely. I didn't have time to look at it.

P02

Is really interesting to hear cause like I was on pens for quite a long time and and then I can't exactly remember how many years. But like the, there became a point where they just stopped working for me like I’d just I'd do an injection and just wasn't effective anymore, which is why I moved over to a pump. And like I found that it was really funny, like because when I moved I was on Omnipod at the time and like suddenly having these like infusions throughout the day rather than like, you know, big old injections, actually felt like. I remember distinctly feeling like I almost wasn't diabetic for the first

time in how many years? And it was like really noticeable, obviously, like been on the pump now for a long, long time, so the novelties worn off. But, yeah, it's interesting to hear like I do think just going back to the whole like how if if you hadn't done pens, I think it would put a lot. I think it would be really important that the somebody who didn't have that experience was able to access like a a healthcare team that could help them, you know, or have some kind of training on how to do it. But obviously in reality that's like, you know, depending on where you are in the in the in the UK, like not every health, diabetes healthcare team is good, unfortunately. Like yeah, like I'm super lucky like you know where I am like I have a really, really good team. And paediatrics was like he actually an amazing team. But there was a time like when I went to uni or when I first transferred into adults like there was just one nurse that well, I ended up, they ended up bringing back like a old diabetic nurse from retirement, like and who was clearly not. I'm saying my opinion here, but she was clearly not fit to be brought back from retirement. Um. And like that massively made my mental health terrible because at the time I needed support and I wasn't getting it, she wasn't even listening to me, she was basically like insulting me. Erm. And I was really lucky to then move to, like, kind of like the a different hospital, but within the same NHS Trust and they had a different team and their team was amazing and I've been with them ever since. But, you know, it's like if you were a person that hadn't had basic training or like, you know, knows how to use pens and in the you know in the emergency like you're on holiday or something, you're pump gets stolen, it's broken or I mean, you haven't got that with you. How, how are you gonna get help in that situation, if you're one of those unfortunate people who've not got a good healthcare team? I think that's, yeah, that would worry me but you know, it's just, yeah, the state of the NHS's lack of funding.

P05

Yeah, cause like I always think it like I always compare going from like a pump and CGM back to pens and finger pricks. It's literally like going from using a supercomputer to counting on your fingers like the downgrade, I think really is that much. And one thing that was like really life changing for me with the CGM, was seeing overnight blood sugars and then being able to use my pump to adjust my basal really quite precisely. Like obviously with the help of my team but now that I've been on the pump for quite so long, I feel confident to be able to like look at, you know, my CGM and go, "OK, I keep getting hypo every night at like 3:00 am, I'm going to reduce my insulin for like the two hours previous to that, to prevent prevent that and hopefully smooth out that line and then we'll see how we go". So, I think having the kind of, the basic diabetes skills of like even being able to like put together your pens and like use proper like sharp safety. Because like these are not things that they're necessarily going to be like teaching kids and their parents who are just newly diagnosed because it's like, "OK, so you might have to do this in an emergency, but this is how you're putting in an infusion set with a pump and blah blah". And yeah, they might have, like, a stockpile of like, insulin pens and needles at home and God forbid they would ever need to use them. But I think as the technology moves on and things become so much more I suppose AI dependent because that's the way that we're going. Like these sort of fundamental skills for, like basic diabetes care are being kind of forgotten and like underutilised because people aren't seeing mid necessarily the value in them. But I think that's like a real disservice to like these kids who are then going to grow up. Because, like, there's a young man that is in my school and he's just going under year 10 now. So, his parents self-fund Dexcom, like he's on a closed-loop system, but obviously it's it's self funded and I'm always very suspicious of even my FreeStyle Libre. Like if I have low blood sugar because sometimes I don't feel low and then I'll check it and it'll I'll be like 5 instead of 3.2 and then I'll check the the CGM again and it's rebounded back up and it's like a flat line, five. So, if I had treated that hypo, I would then have

ended up, you know, maybe 11 or 12. Which isn't the end of the world, but still it would have been unnecessary. And I remember one day like being in lesson with him and I said like “ohh can you just check your blood sugar for me to back up your like the reading on the sensor” and he just went, “I don't even have that anymore”. And I was like, “what?”. And he was like, “I've not had that in years, Miss, like I just used my sensor”. And it was like. It was like being like a baby boomer and coming face to face with like an iPad kid for the first time. I was just like “I have a CGM and I don't leave the house without my blood monitor just in case”. But like I just felt so like outraged cause I was like “who's giving your parents the some information? Because that is like a basic diabetes care thing. Like, I don't leave the house without a glucose source of some form and like my like obviously my pump and stuff is always on like and like I'll usually have like a spare cannula, like floating around in a bag somewhere”. But like, I don't know, I just, I feel like so many people are, like, getting too caught up in the technology and there's not necessarily been a lot of what if this fails conversation for like on a bigger scale. Because like, obviously we're all talking about it individually, but like say, I don't know, for whatever reason, Medtronic came down with a virus. I don't know, all the pumps were hacked. I don't even know if that's possible. But like we were all then collectively screwed, what will we all do? There would have to be a thing for that.

Interviewer

I I wonder if future though do you think or what what do people think? Do you think that's always gonna be the case? Because I guess before you know blood finger prick tests people using like urine dipsticks to manage their blood glucose levels. So, I don't think there be many people now who would say everyone should still know how to do that. Do you think there'll be new technology in future that changes that?

P05

That's actually a really good point.

P10

Well, to be fair, no. I would say like it depends on the accuracy cause urine is less accurate than blood, but blood is still more accurate than the CGM. Once you get to the point where the Dexcom is foolproof, which I don't think it would be because as [P02] said earlier, it's intercellular fluid, it's not blood. So, like if it's perfect, which, like you, couldn't get it to be, then yeah. But I would say the bloods are a more foolproof method whereas like urine, it's still touch and go because it's not the same and it takes longer. But like I'm the same, so I always take a pen and I take finger pricker and all that stuff out the house with me at all times. Like I I wouldn't feel safe about it just because it's a fall back method. But yeah, I would say like it depends on the accuracy of what you're talking about.

P05

Definitely. And I think fundamentally it's like that feeling of safety as well. Like I have like like zip pouch with all my like pump stuff. So, if I need to do like a reservoir change or like I've got batteries in there, I've got like wipes to like take off the sticky if I’m changing a cannula. But like I. Like if I'm going

somewhere overnight, like if I go to work and like my pumps full on whatever and the battery's fine, I don't feel the need to take all that stuff with me. But in like my desk at work I have like a spare cannula, and I'll have you know, like glucose tabs and things like that. But those are all you know, like those like little safety nets that could prevent a bigger catastrophe if something was to crop up. So, like these little things relieve that strain or that potential stress of that happening. But I think the phasing out of like blood glucose testing, I don't think is that likely. Just because, as [P10] said, like even when we're talking about CGM, I still call it my blood sugar even though, which the point that [P02] made earlier, it is interstitial fluid. It is not actually a blood glucose. And I don't see that terminology or like that medical sort of speak changing anytime soon, because fundamentally that is still what we're talking about. We're talking about blood glucose and the management of blood glucose because ultimately, that is what every sort of bolus, basal, whatever, that's what we're all fundamentally collectively aiming to do is to be better diabetics. So, I would love to think that the technology would get to a point where it's so pitch perfect that we wouldn't need anything to just like safeguard or fall back on, but I just don't think that that's the world that we live in, and it's not necessarily a realistic thing to maybe bet on.

P01

I think I think that was like highlighted recently when the when the FreeStyle app like went down. Um. And I saw like a lot of there was a lot of like anger about it on social media, understandably. And it was a it like for me it was a massive inconvenience but I did just go back to fingerpricks for like the immediate time and then back to the old version of the sensor eventually and then when it fixed itself. But like there were people saying on social media that they were like on holiday and they didn't have any spare things or any means of testing with them. And like I I do leave the house without spare stuff like I don't, when I go into work, I don't take a blood metre with me. But I could like I my office is like a 10 minute walk from my house, so if I needed to, I could I know I could go home. If I went anywhere like further than that, if I go on like a day trip somewhere, I'll have like all of that. Same as you like a zip thing of all of that spare stuff. And so, I did find it interesting that people would go abroad or on a holiday without like a sort of backup option. But I guess, like everybody manages it differently and everybody's got different, like levels of trust in it. Yeah. And yeah, I I don't know, I think. I just think that having a a finger prick thing is such a like simple thing to carry around with you. And like the strips like last for ages and it's not something that like expires quickly, so I think it's an easy thing to use as a backup. Whereas if you had to go back to the urine testing like that would just be a whole other level of faff, then it wouldn't be worth it for the like result that you get from it. So, I think it's like, just sort of balancing out between like the effort and the the amount of effort that it would save you. But one thing that I do like about the technology and why I don't carry a fingerprick test with me like literally everywhere is like the size of it. Because one thing that I used to hate was having, was in like school and stuff, having to carry around such a big bag with all of my stuff in. And people would always make comments that I'd always have a bag with me. And I don't know why it's such a like it's such a silly thing, especially like as a woman, when most women have handbags, it's. Like I know my brother, who's also diabetic, found it even more hard as like a teenage boy having to carry around what looks like to his friends a handbag basically. And so the technology allowing you to like leave the house with less stuff on you does make that, like better as well. And I think that takes away some of the the mental load of like thinking through all the different things you have to have with you and the sort of just the the stress of like, “oh, are people going to, like judge me for this” or whatever. Because it's all more discrete, there's no like sort of having to do an injection or a fingerprick in public which draws more attention. But yeah, I think. So I

think if you had to carry around spares of everything all the time, it would almost like to take away some of the advantages of the technology for balance.

P05

It almost defeats the purpose, doesn't it?

P01

Yeah, yeah, exactly. And I think it's finding the balance of like, personally, if I went on a holiday abroad, I would definitely take spares with me. And I would leave them in the hotel and that would be like that would be fine if I know I could access them if I needed to. But I wouldn't if I'm like, popping to the supermarket, I wouldn't carry all of my diabetes stuff with me.

Interviewer

Yeah. Like [P16] actually. Do you have any opinions on this or what, what do you tend to do?

P16

Um. So I tend to just take my metre and then like a hypo treatment everywhere. But I think it's also like a force of habit as well as a safety thing because I've just done it for 10 years. So, like now I just do it kind of without thinking if that makes sense.

Interviewer

Yeah, no, definitely. Is that something, something you think you're always going to do or do you think will ever change in future?

P16

I think. For the near future, I'll probably just still do it always, especially because recently I've had more problems with my sensors being inaccurate or not working so. But then I guess if things got better then, and technology kind of improved then maybe I'd probably not do it.

Interviewer

Yeah, definitely. And I think that's like, to move things on a little, just purely in the, yeah, the interest of time, that's something I do want to kind of get people's thoughts on as well is the future of technology and where they think these changes will go. So, what do people think is going to be that next advancement we see maybe from a closed-loop or otherwise?

P10

I would almost say like I know with t:slim, um the next update that I'm very keen to be able to do is change the target setting because at the moment I believe it's fixed at 6.5 or something like that as like the base reading. And that's one of the things I found most difficult moving from pen to pump because the pen you can aim to go wherever you like. But with pump it's still, because they're so hyper focused on safety because it is a relatively new thing, they're like leaving massive margins of error. I would preferably like to run at 4.5 all day, like that's never going to happen, but that's what I'd aim for because that's the kind of level I feel most comfortable with. So, I think in the near future, I think pumps being able to do more like freedom with what you want to do as the the technology gets more reliable. And as companies kind of get more reassured that if they let people choose, people aren't gonna start dying and they're gonna get sued and stuff like that. It's something it seems to be like it's they're overly cautious I think at the moment and as it gets used more. It's not even really an advancement in technology it's just an advancement in how they're regulating it I think is going to be the next major thing.

P01

That's interesting because I think I'm, I'm pretty sure on my pump I can set my target is whatever I like so.

P05

Same.

P01

So, are you in the Medtronic pump as well [P05]?

P05

Yeah. And like that's cause. That kind of shocked me that it was so set like my like glucose range is. Um. I have mine sitting at 5.6 to like 7.8. So, like ideally if I was like 6.5 all day, I would be like this is the best day ever, whereas like obviously for you like 4.5 is like you're most comfortable. Whereas like I would feel like on the brink of a hypo all day if I was that like at that level, but obviously like it's such an individualised condition as well. But the Medtronic one, like there's no limit on what you can set. Like it does give you a little alert when you edit it because. I think it's like 5 point whatever to 7.5 or something. And then if you edit it's like um "this goes beyond the like recommended professional blah blah blah" and you just have to click OK and then you can save it. I don't know if there's a safeguarding mechanism in there, like if you wanted to put your blood sugar as like something I don't know crazy as like 2.5 to 4 or whatever. I don't know if it would allow you to do that. I don't see why anyone would want to. But yeah, I didn't realise that that was a thing with the t:slim that you couldn't change it.

P10

Yeah, that is only with the Control IQ because basically any time it automatically corrects, it won't let you automatically correct for it will always automatically correct 6.5, but like, that's still something yeah, I found annoying.

P01

Well, I think that kind of shows that I mean for me, one of the most helpful things that like these companies could do would be to have like more collaboration, which is never going to happen because they're competing companies and they're all trying to make money. But all of the different systems all have their like positives and negatives, and it's really frustrating having to choose a compromise. I mean often, obviously you you don't really get to choose because it's a bit of a post code lottery of what is funded in your area. But if like I could choose between everything, I would still feel that I was like having to compromise certain aspects. And I wish that there was some kind of way of these different technologies being like compatible together, like different pumps with different sensors. Or like just they were allowed to share their designs more because it's really frustrating that like healthcare and sort of people's qualities of lives that are just basically dependent on the profits of these companies. But I get it, they're businesses at the end of the day, but as of like, I don't know, I guess cause we're we're customers, not patients of of the companies and and that's the difference really.

P02

I think it's on that point. It's slightly I think a lot of these companies aren't based in Europe. They're like US, aren't they? They're based in the US. So then in the US, they're paying customers, but it's different in the UK cause it's the NHS as the customer and then they get the funding. And so it's like yeah. Because I had like. That's the one downside to the thing I've had with the Dexcom sensor is, I had it. I've had it for over a year now and when I first got it, I had no problems with it, but then sort of around September last year I literally started getting like contact dermatitis from the glue.

P01

Oh yeah, I think they, didn't they change the adhesive or something? Wasn't there some kind of thing about that?

P02

Yeah, they changed they changed an ingredient in the adhesive which. Because I looked it up and I mean, I'm also on some other skin medications, so I have quite sensitive skin. But given that I was completely fine like prior. And then it was changed to the point where like under the adhesive, I literally had like blisters and like sorry excuse the grossness, like puss and like it was horrible. And then the it scarred for about like six weeks it wouldn't go away like until it healed properly. And and I was like on the phone to them being like, you know, and it they were just like, “ohh, we're really sorry but you know, when we changed the ingredients like we always expect, some people are gonna react badly to it”. And it's just like, what kind of explanation is that? Yeah. And I I called them up and

they sort of. I spoke to someone who, well, by the time they had, like, made an appointment for me to talk to a specialist, I'd kind of already figured out, like done my own research and found like a solution to the problem. Like cause now I have to wear like hydrocolloid plasters underneath and then I ordered this like plastic thing from the US that goes underneath and then I have to stick it all down. That's, it's like, you know, I'm spending like extra money to be able to wear this thing, but it's worth it because it, you know. But it shouldn't really be like that, because there's also evidence like, you know, I found it on the internet, evidence to suggest that the ingredient they switched it in with was already known as like a dermal dermalogic, like a react skin reactant. So, it's like why did you bother changing it? Probably to make save money like yeah, so. I guess, yeah, that's frustrating. But I think just going back to [Interviewer]'s kind of question about, you know, where is technology going. I think, I think to be honest, I I think it will keep advancing and I I do think like the integration of activity data or how like people's energy expenditure into these like algorithms, AI technology will be the next step, until. And I think it will keep, technology will keep developing overtime, I mean it'll they'll always be the pain in the arse of waiting around for it to be funded, at least here. Yeah. So, we'll be, you know, on that time lag and what have you. But I think it will keep going in that way until they find a way to mask beta cells that you can put in in the pancreas. Because that's until there's like, as I say, a way to to have a body do it on its own or to, you know, find that cure or a way to mask, beta cells and have them survive in the body then, then the technology will become redundant in that or until you know people can have that kind of treatment on a regular basis. But yeah, I think, yeah, until it's redundant, I think it can keep going. Or maybe it will reach a plateau at some point, but yeah, that's I always think.

Interviewer

Yeah. Sorry, I was just gonna say like. Yeah. In terms of feeding that extra information and what what do you think it's gonna help with most or what times do you think it can really make that difference?

P10

Just going to say like because I was obviously on the as I said before, a trial where they tried to extend my honeymoon period and it was kind of it wasn't tech. It was just an injection every three months but I think it really helped because at least for me, it seemed to have worked. I don't know if I was actually on the drug because it's just double-blind and stuff like that, and it's still like the results waiting to be in for like a few years and stuff. But I actually found there was a brief period of time where I was off like insulin, like I only had Levamir and I didn't I could just eat normally. Like albeit I was having like 30 grammes of carbs at a time, but like it helps so much and just kind of learning easing into it like that. So, I think that's kind of it's tech, in a way. It's not kind of like technology, but kind of medication, that was a really interesting thing because it let me have more time to adjust. So, though it's not a cure by any means and I had to go off it because the study ended, I think that was quite an interesting way to kind of ease newly diagnosis into it without kind of as much of danger. So, I think that's gonna be quite a big advancement. Because I did an EPQ into this stuff, so I've like read into it little and I know that is something they're like trying to do, kind of extend the honeymoon period, cause it's quite a quite an easier option for now.

P02

Yeah. It's like identifying people before they get to the point of us like oldies, who've had it for ages. Like they don't want us on trials because we're useless. Yeah, but yeah, I remember yeah doing a lecture, she does a lot of work for, like, JDRF, but she I remember going on like one of her units and lectures about diabetes and them trying to identify like people or like kids pre diabetic and like using the antibodies and stuff to try and do that kind of thing. So yeah I think technology will keep, keep going until it's redundant, I think. Yeah.

P05

Do you think that's going to be a thing in our lifetime?

P01

I think we'll all be old, I think.

P05

We're all gonna be old. I mean, God, I feel it now, but, like, especially like I've been. It's weird when you get to the point where, like, you've lived longer as a diabetic than you lived as a non-diabetic. It's like that weird sort of the balance shifts in your life, and then you're like, "oh, God, this is just going to be me forever now". But like where we're at now with like the closed-loop system like I honestly, I am like in awe of the technology that we have available. Because I remember just like it was so out of reach whenever I was like newly diagnosed and you know in the space of, you know, 15/16 years, we've seen these massive leaps and bounds with like wearable technology, CGM, pumps have become more accessible. And now we've got the integration of AI but like. Maybe maybe this is me thinking small mindedly or like thinking like not thinking big enough, but I just think at the minute like the closed-loop CGM system seems like at the very forefront. And God knows what, like the engineers are doing in the background like I know they're always like tinkering away and like there will always be like the next big thing, whatever that might end up being. But, I would love to think that they would be able to do like transplants and like cure it at some point. And like you know, maybe you know another 15 years, it's a long time but like in the grand scheme of like medical, medical technology and breakthrough and like our lives, like we're not going to be that old like. So, it could be trialled, at least in humans, at some point in our lifetime, in the next like decade and half, which is a really crazy thing. Because like I remember for like so long when I was firstly diagnosed, like being in such disbelief that it had happened to me being so young and being like, "Oh my God, my life is fundamentally changed forever. I'm never going to be the same." And now like diabetes has been such a big part of my life for so long. If someone like came along and was like "ohh by the way, we're gonna try all this surgery and you just might not be diabetic anymore", I find that I think that would be a really. I mean, I'd love to say that I'd jump at the chance, but I don't know because I think the thought of like getting my hopes up and like maybe being like, "oh, I could wake up and get out of the hospital in two or three days and never be diabetic again". To then maybe the surgery failed, or it didn't work or something like. I don't know if I could put myself through that from like, a mental health perspective because it's almost like woefully opening yourself up to like a form of grief like it's, I don't know. It's a really weird thing to think about. Like am I the only person that's ever had that thought?

P02

No, I I think I've had similar because I felt for. I'm I'm, you know, I think I got. I was diagnosed when I was 13, which was like a pretty awful time to get it. Um. And then like, similar to you, I just took on a lot of my care. And then I don't think. I think when I was a teenager, like sort of, I went from child to adult very, very quickly and then maybe do like kid stuff like in the way they do when you're a teenager and and found uni quite stressful. And it's not until like I think just looking after my health alongside trying to live your life is very difficult. And like I've still battle with the kind of thing of like diabetes is not my personality, oh, I I personally don't want it to be. Like some people make it like if they've got a long-term condition or like a learning disability or anything, like they'll make that part of them. But I prefer to have it not as my identity. But it's very hard to like separate that because then at times I felt it has really stopped me from doing things I want to do. And it's not until now and I'm kind of done a lot of personal work and had some like therapy and things to help with other parts of my life. It's not until like now and I've done difficult things and what, I'm 25 this year like and I can sit down and say “right, OK, like, actually, I'm gonna prove myself or prove my worst fears wrong and I'm gonna go and do the things I wanna do, like travelling abroad for like 8 weeks” or whatever. Like, do the things I actually want to do and not let the diabetes stop me. So, I think if you know, if somebody did turn around and go “ohh, you can have a cure”, I would say yes. Like and I think. I'm I think probably on the borderline between optimist versus pessimist. Like I I do. You know, I. But I'd probably be willing to try it. Um. But I do, like recognise at the same time, I think it would be very weird to suddenly wake up and not have all of that and but also know that actually you've gone through like. I suppose if we're what, 50/45, by the time. I don't know. I don't know how long it would take, but you know, knowing that you've got to the point of your life when, it sounds really biassed, but you've had the best years of your life, like the young years of your life, and now your diabetes free. It's kind of like, it will be a whole other load of grief or like a whole other load of processing, I think, yeah. I guess I for me, like I think. The mental mental side of diabetes is like probably the hard or equally as hard as the physical.

P01

I think also like for me, if the even if there was like a cure literally tomorrow and I think it it well, diabetes would still like affect my day-to-day life. I think just going through that like diagnosis and then living with it for so long has like irreversibly sort of affected my mental health and changed how I think about like myself and how I organise myself and how I think about my body as well. And I don't think. I think even if I was like cured, I think it would take a lot of like working through things to sort of. I don't. I just don't think I would get back to how I was before. I think it like it does just for me it introduced like a lot of anxiety. And just the fact that you have to think about everything in so much detail, I think it would be really, really hard for me to like, train myself out of that. And I suppose that, like similar with with the technology, if if the technology got to a point where it literally did everything for you, I think it would take a long time of using that and trusting it for me to get out of the sort of overthinking that diabetes has made me do. And because that's like sort of seeped into every aspect of my life as well, because you have to think in so much detail about this one thing, it means that I think in so much detail about. I don't know, like if if I'm going somewhere other things I have to pack and take with me, or like the food that I'm eating, it really, really, I think makes you think a lot about that. Um. And it would just be really weird not to have to do that. So, I I don't think

it would be as simple as like ohh we've got this solution and and then all your sort of worries are taken away.

Interviewer

That's really good. Really interesting to hear.